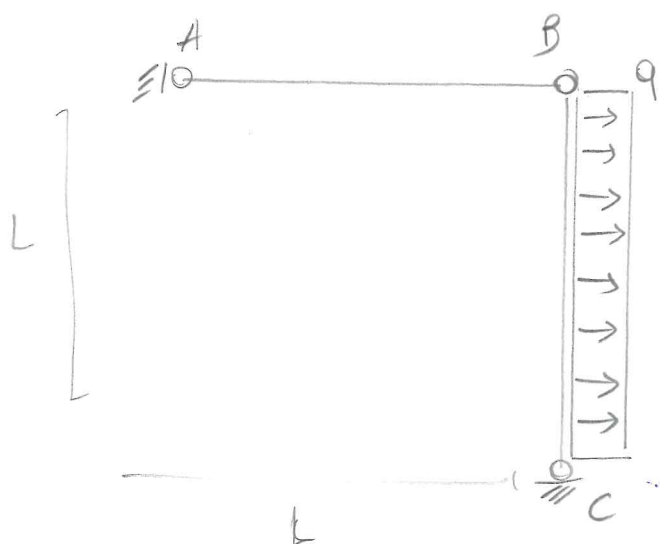


TLV ISOSTATICO:



? φ_C

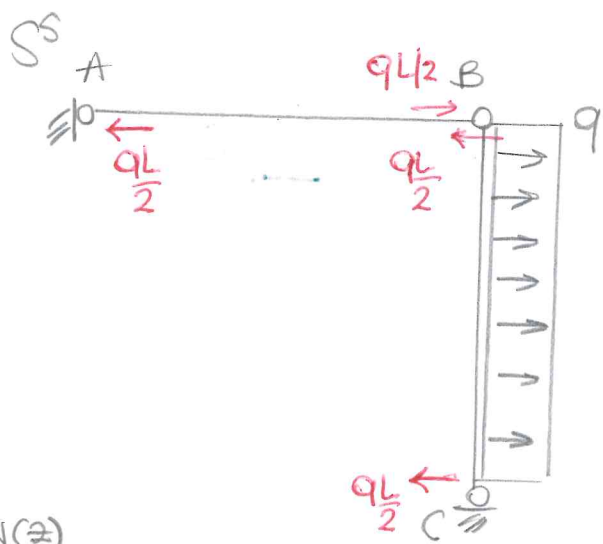
$$3t - s = l - 1$$

$$6 - 6 = l - 1 \quad l = 1$$

$$\nexists C \quad l=0 \quad 1=0 \rightarrow \text{ISOSTATICA}$$

→ TLV

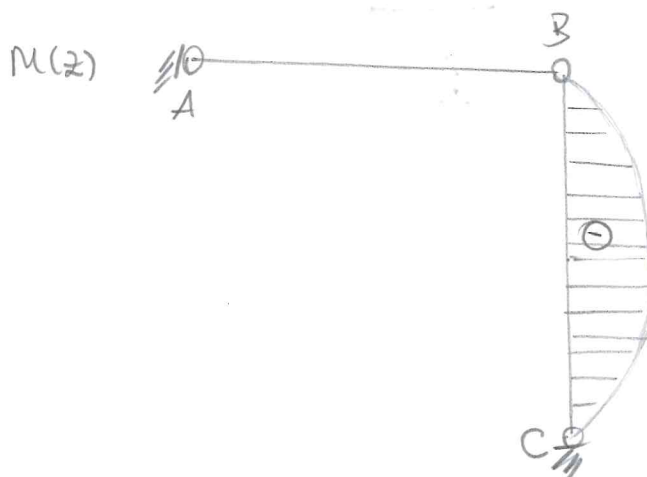
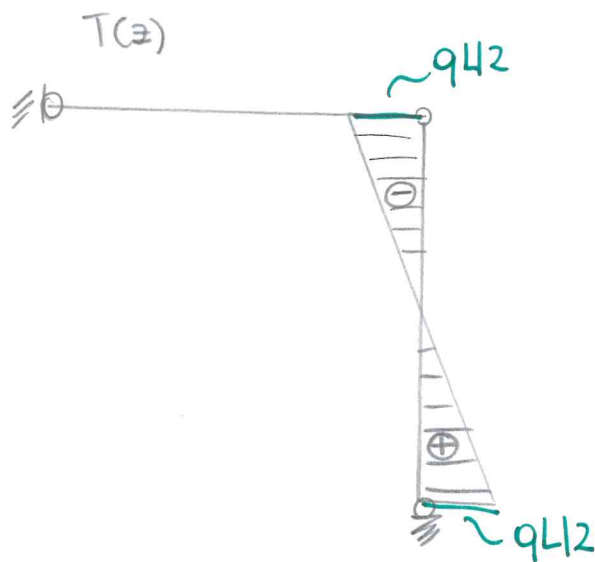
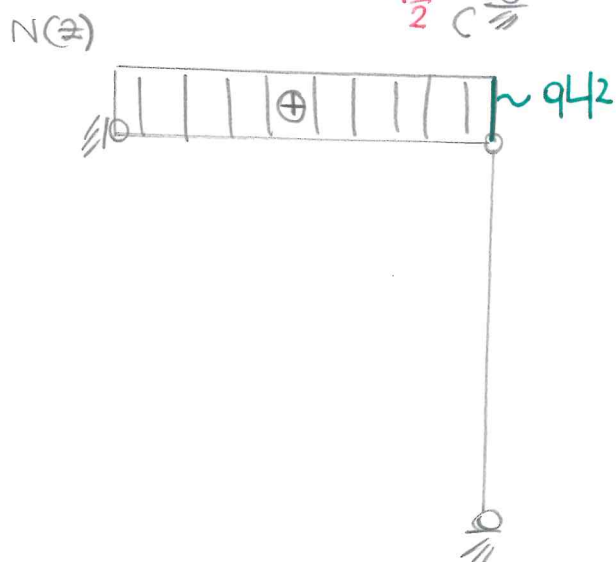
$$L_{ve} = L_{vi}$$



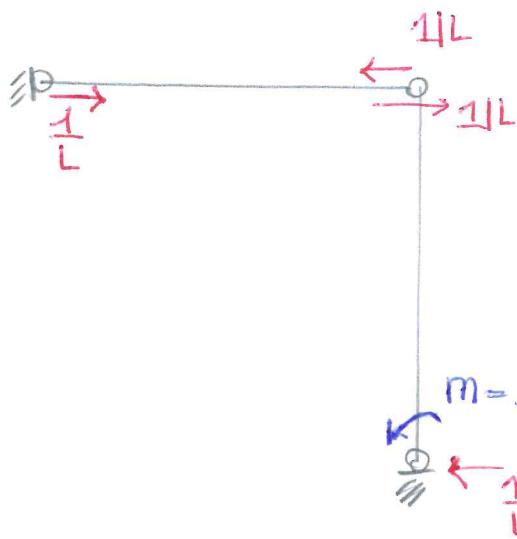
$$N(x) = \begin{cases} qL/2 & \text{in AB} \\ 0 & \text{in BC} \end{cases}$$

$$T(x) = \begin{cases} 0 & \text{in AB} \\ -qL/2 + qx & \text{in BC} \end{cases}$$

$$M(x) = \begin{cases} 0 & \text{in AB} \\ -qL/2 x + q \frac{x^2}{2} & \text{in BC} \end{cases}$$



8F

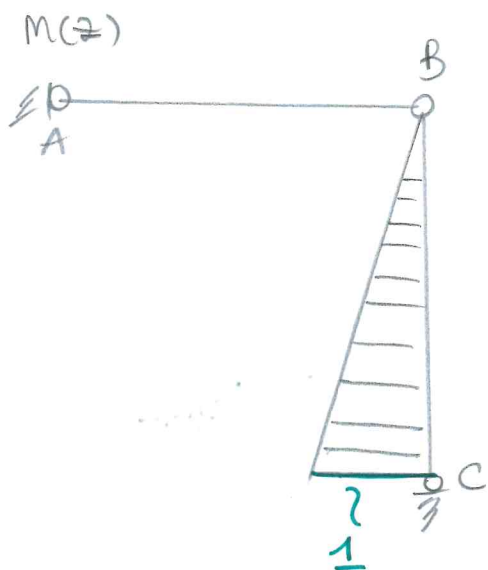
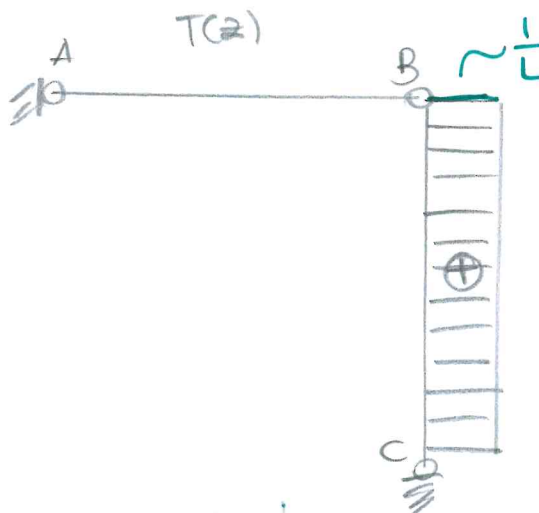
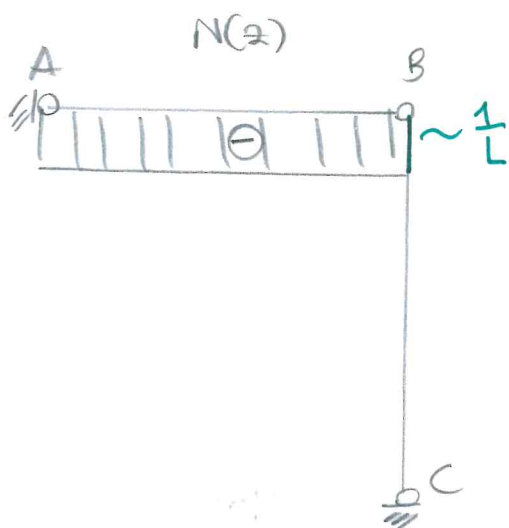


$$N(z) = \begin{cases} -\frac{1}{L} & \text{su AB} \\ 0 & \text{su BC} \end{cases}$$

$$T(z) = \begin{cases} 0 & \text{su AB} \\ \frac{1}{L} & \text{su BC} \end{cases}$$

$m=1 \rightarrow$ Ente unitario
allo spostamento
cercato

$$M(z) = \begin{cases} 0 & \text{su AB} \\ \frac{1}{L} \cdot z & \text{su BC} \end{cases}$$



$$L_{ve} = L_{vi}$$

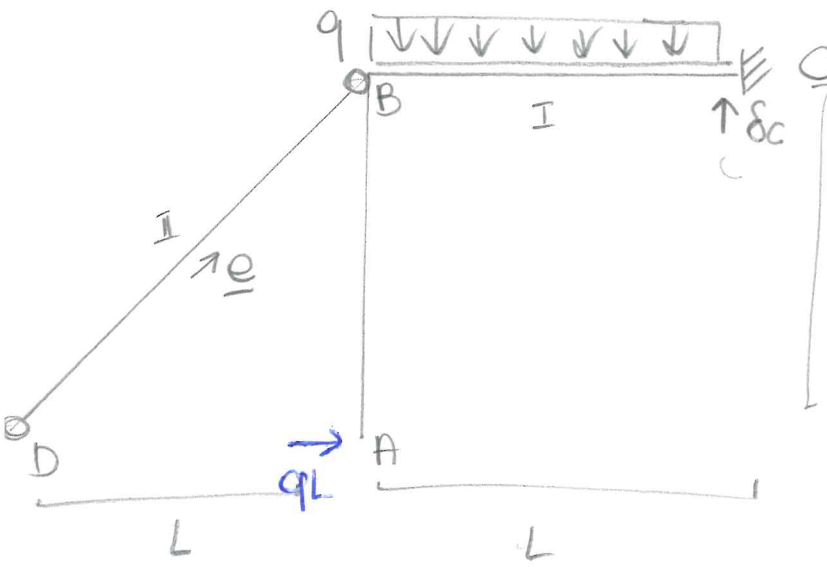
$$L_{ve} = 1 \cdot \varphi_C$$

$$L_{vi} = \int M^F \frac{M^S}{EI} dz = \int_A^B \cancel{M_{AB}^F} \cdot \frac{M_{AB}^S}{EI} dz + \int_B^C M_{BC}^F \cdot \frac{M_{BC}^S}{EI} dz$$

$$\varphi_C = \frac{1}{EI} \int_0^L \frac{z}{L} \cdot \left(\frac{qz}{2} (z-L) \right) dz \rightarrow \varphi_C = -\frac{qL^3}{24EI}$$

$$\varphi_C = -\frac{qL^3}{24EI}$$

TLV IPERSTATICO:



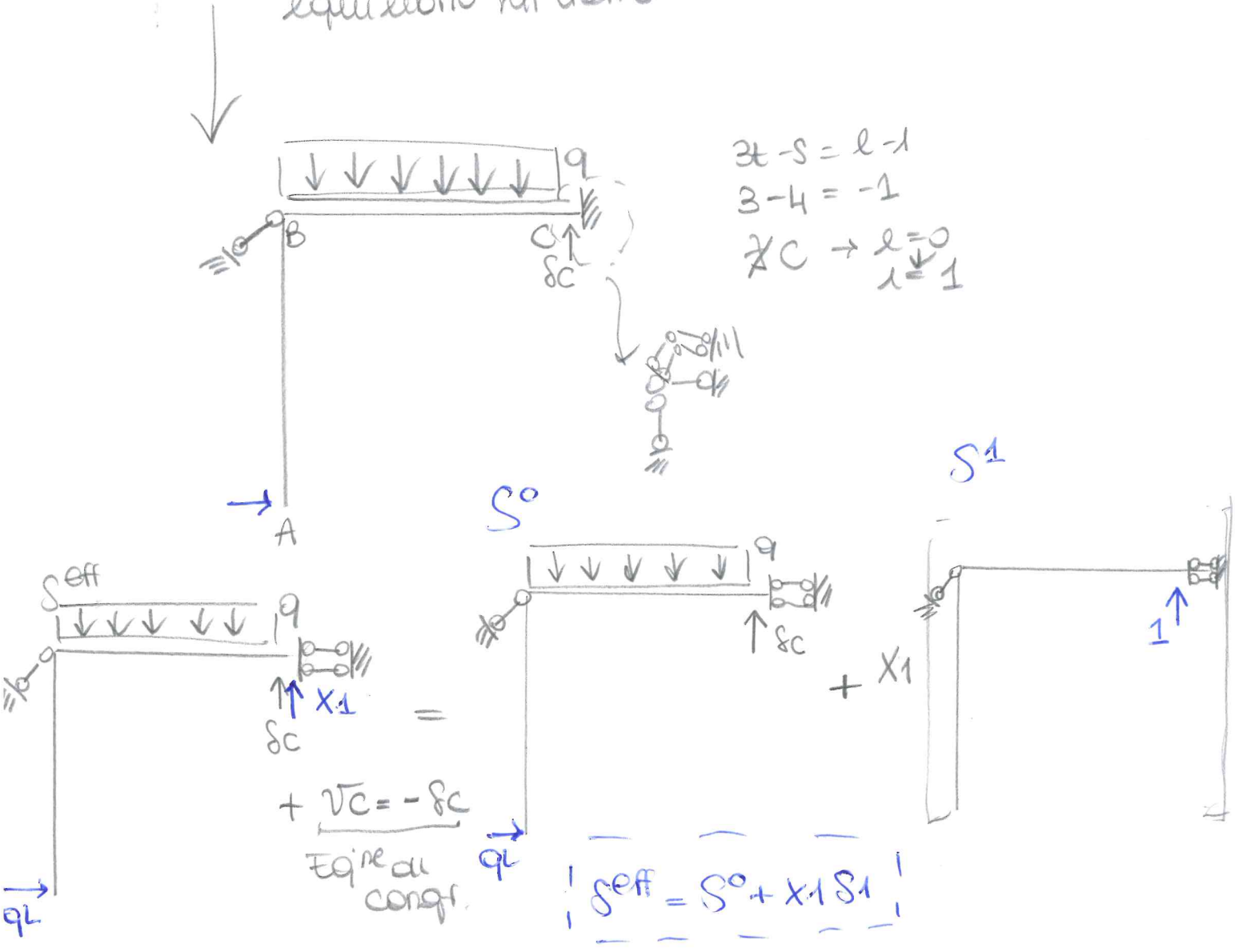
(BD) II: lo riguriamo come un penolo esterno incernato a II/II

- corrispondenza (+ Hp EA -> infinity) Cinematica!
- (S_B · S_D) · e = 0 -> S_B · e = 0 (cerniera)
- corrispondenza statica!

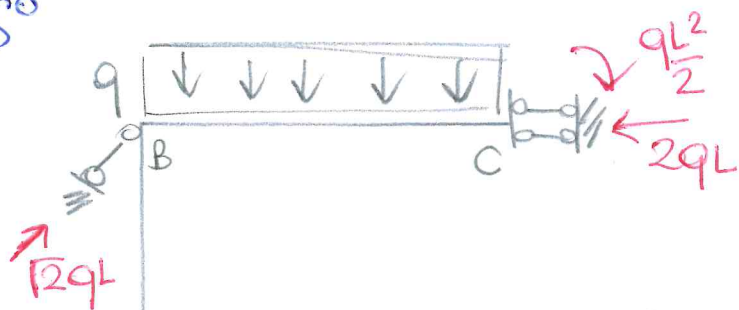
In assenza di azioni esterne concentrate o distribuite su BD le reazioni sono dirette //

ostacoli non ci sarebbe equilibrio sul tratto

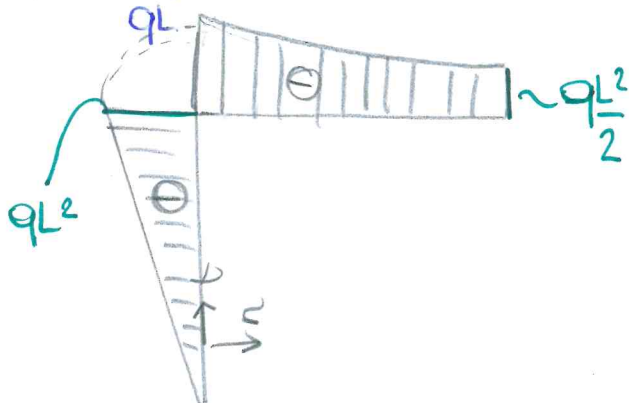
del tratto BD



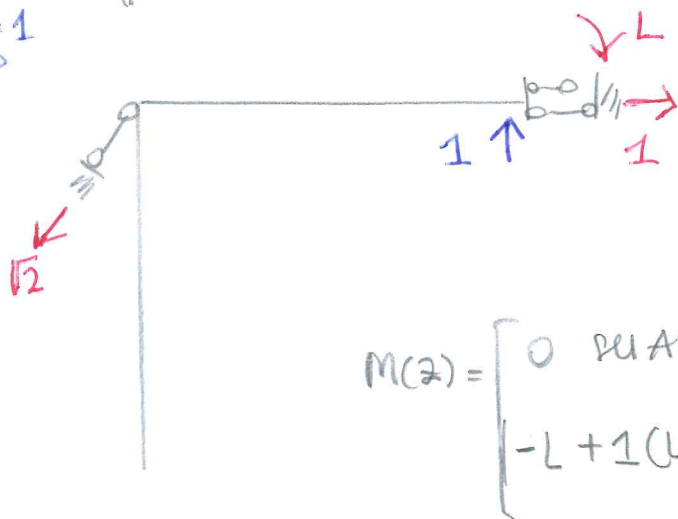
S^0



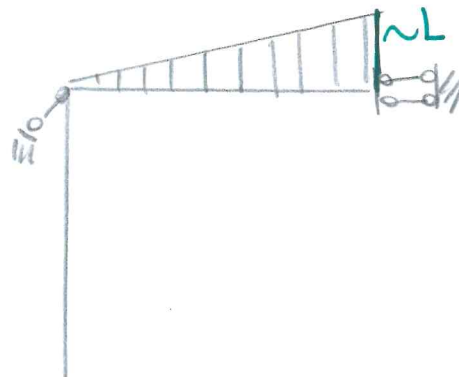
$$M(z) = \begin{cases} -qLz & \text{on } AB \\ -\frac{qL^2}{2} - q\frac{(L-z)^2}{2} & \text{on } BC \end{cases}$$



S^1



$$M(z) = \begin{cases} 0 & \text{on } AB \\ -L + 1(L-z) & \text{on } BC \end{cases}$$



$$Lve = Lvi$$

$$S^S = S^{eff} = S^0 + X_1 S^1 \quad S^F = S^1$$

$$Lve = -1^F \cdot (-\delta_C)^S = 1 \cdot \delta_C$$

$$Lvi = \int \frac{M^F \cdot M^S}{EI} dz = \int_A^B \frac{M_{AB}^F \cdot M_{AB}^S}{EI} dz + \int_B^C \frac{M_{BC}^F \cdot M_{BC}^S}{EI} dz$$

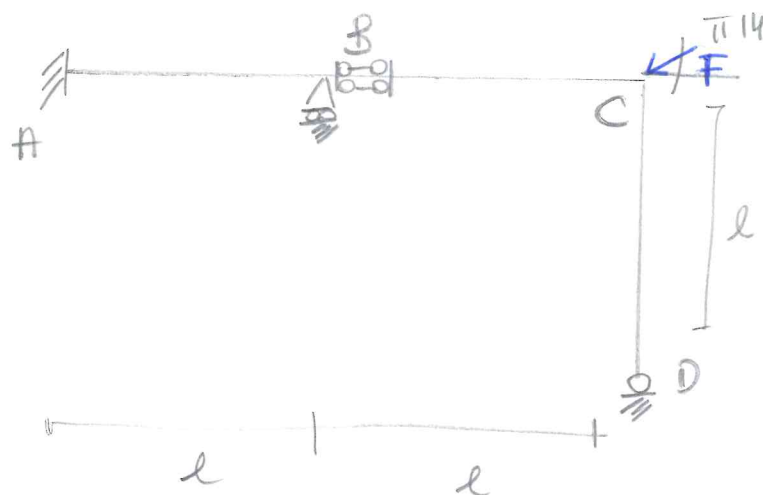
$$Lvi = \int_0^L \frac{M_{BC}^F \cdot (M_{BC}^0 + X_1 M_{BC}^1)}{EI} dz \rightarrow \int_0^L \frac{-\frac{z}{EI} \left(-\frac{qL^2}{2} - \frac{q(L-z)^2}{2} - X_1 z \right)}{EI} dz$$

$\underbrace{-\frac{qL^2}{2} - \frac{q(L-z)^2}{2}}_{M_{BC}^0} \quad \underbrace{-X_1 z}_{M_{BC}^1}$

$$Lve = Lvi$$

$\hookrightarrow (X_1)$

TLV 2 VOUE IPERSTATICO!



? Risolvere l'iperstaticita'

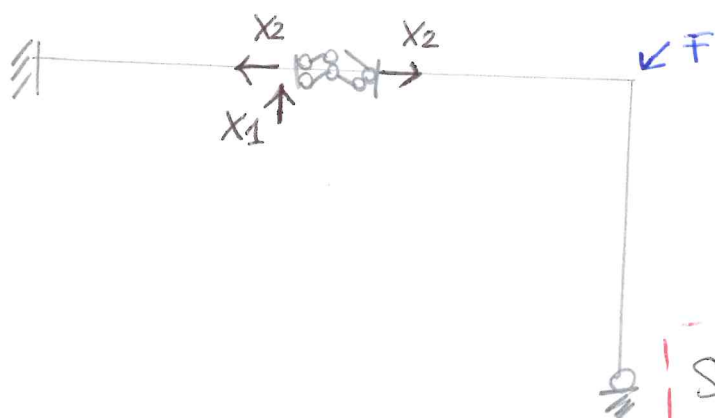
$$3t - s = l - i$$

$$t = 2 \quad s = \sum MS = 3 + 1 + 2 + 2 = 8$$

$$6 - 8 = l - i \rightarrow l = 1 - 2$$

$$\text{At } C \quad l = 0 \quad i = 2$$

S^{eq}

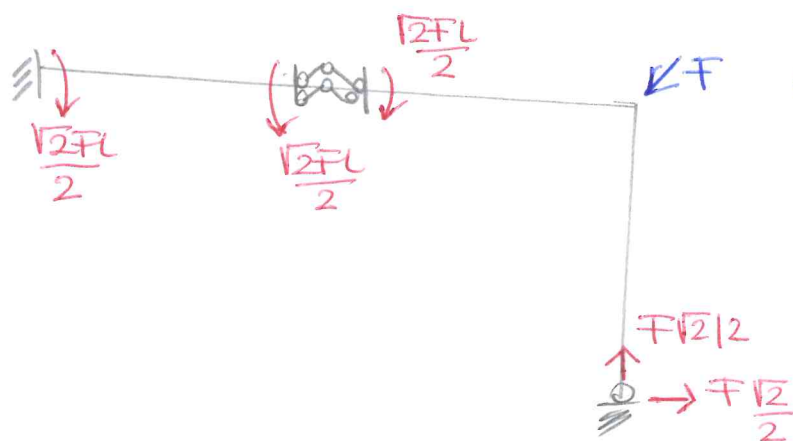


+ Eq'n di congruenza

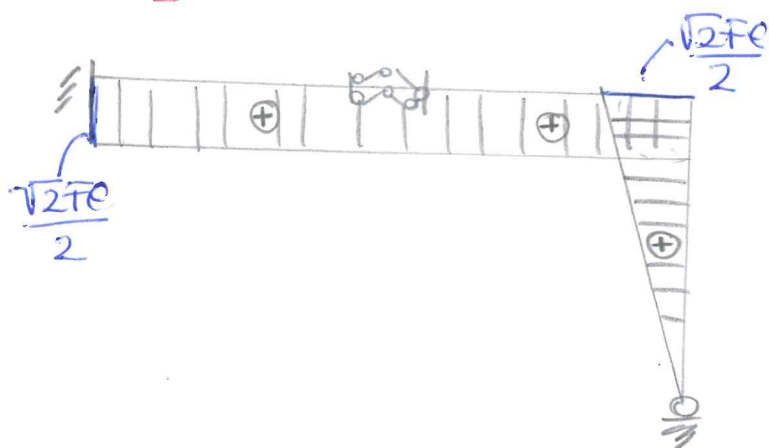
$$\begin{cases} \delta B = 0 \\ \Delta WB = 0 \end{cases}$$

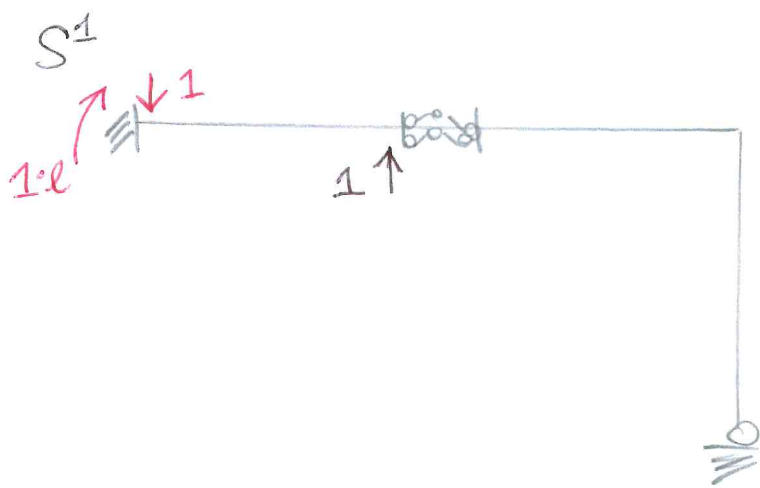
$$S^{eq} = S^0 + X_1 S^1 + X_2 S^2$$

S⁰

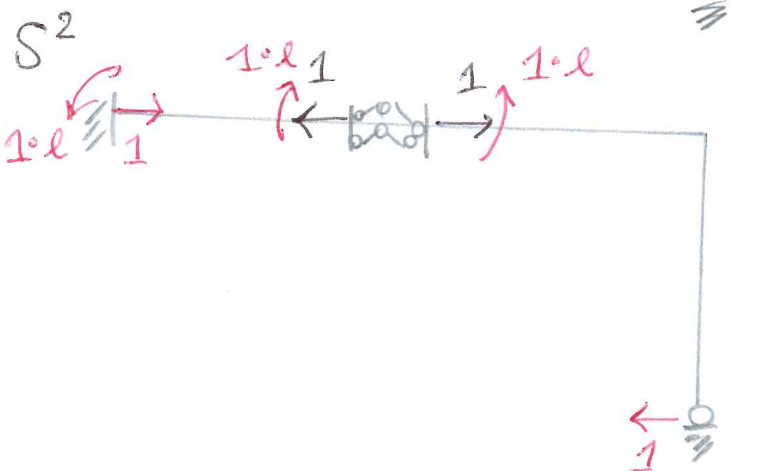
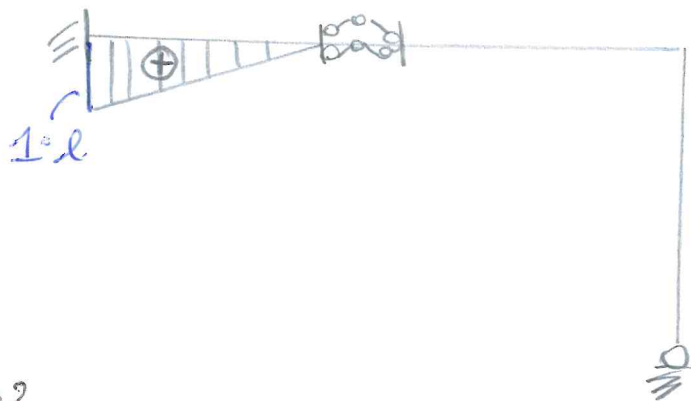


$$M(x) = \begin{cases} \frac{\sqrt{2}FL}{2} & \text{su AB pr} \\ \frac{\sqrt{2}FL}{2} & \text{su BC pr} \\ \frac{F\sqrt{2}}{2} \cdot z & \text{su CD pr} \end{cases}$$





$$M(z) = \begin{bmatrix} 1 \cdot L - 1 \cdot z & \text{su AB} \\ 0 & \text{su BC} \\ 0 & \text{su CD} \end{bmatrix}$$



$$M(z) = \begin{bmatrix} -1 \cdot L & \text{su AB} \\ -1 \cdot L & \text{su BC} \\ -1 \cdot z & \text{su CD} \end{bmatrix}$$



Prima Applicazione del TLV $S^S = S^{eq} \quad S^F = S^1$

Lve $1 \cdot \sqrt{B} = 0$ per la congr.

$$\begin{aligned} Lvi &= \int_A^B M_{AB}^F \cdot \frac{M_{AB}^S}{EI} dz + \int_B^C M_{BC}^F \cdot \frac{M_{BC}^S}{EI} dz + \int_C^D M_{CD}^F \cdot \frac{M_{CD}^S}{EI} dz \\ &= \int_A^B \left(\frac{(L-z)(\sqrt{2}FL + X_1(L-z) + X_2(-L))}{EI} \right) dz \quad \downarrow \quad (Lve=0) \end{aligned}$$

$$Lvi = \int_0^L \frac{\sqrt{2} FL}{2EI} (L-z) dz + X_1 \int_0^L \frac{(L-z)^2}{EI} dz + X_2 \int_0^L -\frac{L(L-z)}{EI} dz = 0$$

Seconda Applicazione del TV $S^S = \int eq$ $S^f = S^z$

Lve $1^\circ \Delta W_B = 0$ per la congr.

$$Lvi = \int_A^B M_{AB}^f \cdot \frac{M_{AB}^S}{EI} dz + \int_B^C M_{BC}^f \cdot \frac{M_{BC}^S}{EI} dz + \int_C^D M_{CD}^f \cdot \frac{M_{CD}^S}{EI} dz = 0 \quad (we=0)$$

$$Lvi = \int_0^L \frac{-L}{EI} \left(\frac{\sqrt{2} FL}{2} + X_1 (L-z) + X_2 (-L) \right) dz + \int_0^L \frac{-L}{EI} \left(\frac{\sqrt{2} FL}{2} + X_1 \cdot 0 + X_2 (-L) \right) dz + \int_0^L \frac{-z}{EI} \left(\frac{F\sqrt{2}}{2} \cdot z + X_1 \cdot 0 + X_2 (-z) \right) dz = 0$$

due Eqⁿⁱ in 2 incognite X_1 e X_2

$$\frac{1}{E} \int_0^L \begin{bmatrix} M_1^2 & M_1 M_2 \\ M_2 M_1 & M_2^2 \end{bmatrix} dz \underbrace{\begin{pmatrix} X_1 \\ X_2 \end{pmatrix}}_{\substack{\underline{X} \\ \text{incognite} \\ \text{ipert.}}} = \underbrace{\begin{pmatrix} -\frac{1}{EI} \int_0^L M_1 M_0 dz \\ -\frac{1}{EI} \int_0^L M_2 M_0 dz \end{pmatrix}}_{\substack{\underline{b_0} \\ \text{vettore termini} \\ \text{noti}}}$$

Matrice $\underline{F}$
ou Flessibilità

